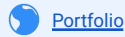


Brian Hinkle

C++ Engineer | 8 Year Game Developer | 1 Shipped Game Title (PS5)



[Portfolio](#)



[LinkedIn](#)



[GitHub](#)



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(704) 604 9356

EDUCATION

BS in Computer Science - Charleston Southern University | GPA: 3.5 Cum Laude

May 2022

Studies: Data Structures, Operating Systems, Architecture, Networking, Linear Algebra, Algorithms, Competitive Programming

PROFESSIONAL EXPERIENCE

Shipped Towers of Aghasba (Editors' Choice Indies #1) - Full-Time Engineer [Video](#)

Apr 2023 - May 2025

- **Pre/Post-Launch Experience:** Met deadlines for milestone builds (trailers, open playtest, Sony certification).
 - Cleared Sony certification by addressing technical requirements checklist and key memory concerns via PS5 dev kit.
 - Practiced proper live player-base coding practices, such as save progress preservation throughout patches/updates.
 - Provided engineer insights to core team meeting discussions (design, tech animation, combat, multiplayer, engineering).
- **Gameplay:** Experience developing for open-world game genre.
 - Created multiplayer-ready systems & features, supporting performant open-world gameplay for players.
- **Multiplayer:** Refactored existing systems for networked gameplay, developing for prediction and rollback in the process.
 - Led solution for client-side prediction & re-simulation for player movement.
 - Established implementation patterns, guided team adoption, and coordinated engineer efforts.
 - Minimized network latency and resolved issues within existing systems, bringing to production ready state.
- **Self-driven efforts:** Performance improvements, dependency minimization, logic nativization, debug tools, documentation.
 - Diagnosed & resolved issues immediately during playtests and improved QA iteration with new workflows.

PROJECTS

C++ Utilities [Video](#)

Nov 2025 - Jan 2026

- Wrote a zero-cost, compile-time abstraction layer for containers. [github](#)
- Wrote a zero-cost, compile-time data wrapper for customized access behavior. [github](#)
- Wrote template metaprogramming utilities for improved language workflows. [github](#)
- Wrote a generalized cache friendly handle system, promoting data-oriented design practices for improved CPU performance.

Game Utilities ([github](#)) [Video](#)

Jun 2021 - Jun 2022

- Wrote a size-generic bitfield abstraction for gameplay tag driven bit packing.
- Implemented shape-agnostic traces capable of exit hits, penetration, ricochets, and strength. Improved performance up to 40% via custom query-area minimization algorithm.
- Developed utilities for cast safety, eliminating silent undefined behavior.
- Wrote utilities promoting the usage of inline storage.
- Implemented print-to-screen tools, using temporary lifetime for code execution with the named parameter idiom pattern.
- Wrote other common utilities (math, collision, debugging, string, logging, materials).

AbilitySystemSetup ([github](#)) [Video](#)

Jul 2020 - Aug 2022

- Wrote fully decoupled and modular input management solution, with support across game/plugin modules. [github](#)
- Developed a more flexible solution for ability input binding, improving workflows and unlocking new capabilities.
- Developed portable avatar actor extension system, properly initializing/uninitializing avatars regardless of type or location.

Multiplayer Web Framework ([github](#)) [Video](#)

Feb 2021 - Sep 2021

- Created engine-agnostic solution for dedicated server management, supporting multiple host machines and game instances.
- Used during development of personal multiplayer game projects.

SKILLS

Languages: C++ (Standard & Unreal), Template Metaprogramming, C#, Blueprint Visual Scripting, HLSL/GLSL, UE Material Graph

Software: Unreal Engine, O3DE, Unity, Visual Studio, Perforce, CMake, Git and GitHub, Jira, BugSplat, Jenkins, Blender, Gimp

Concepts: 3D Math, Networking, Client-side Prediction, CPU Cache, Data-Oriented Design, Object-Oriented Programming, Profiling, Unit Tests, Debugging, Source Control, PS5 Dev Kit, PlayStation SDK, Wwise SDK, UE Gameplay Ability System